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Exp No: 11	*Routing at Network Layer - Static Routing and RIP*

AIM:

To simulate static routing and the RIP dynamic routing protocol configuration using Cisco Packet Tracer.

ALGORITHM / PROCEDURE:

(a) Static routing

1. Place 3 routers (2911 with HWIC-2T serial card) and 2 PCs. Connect PC0 to R1 and PC1 to R3 with a crossover cable, and connect the routers using ***serial DCE*** cables (R1-R2 and R2-R3).
2. Assign IP addresses to all the router interfaces and the PCs as shown in the address table. Set the default gateways (192.168.1.1 and 192.168.2.1).
3. On the DCE side of each serial link give clock rate 64000 and use no shutdown on all interfaces.
4. Ping from PC0 to PC1. It fails because the routers do not know the remote networks.
5. On each router add a static route for every network that is not directly connected: ip route <network> <mask> <next-hop>
6. Verify with show ip route and ping PC0 to PC1 again. Use tracert to see the path.

(b) RIP (Routing Information Protocol)

1. Remove the static routes using no ip route ... (or build the same topology again).
2. On every router enter RIP configuration mode with router rip and version 2.
3. Advertise the directly connected networks using the network <address> command and disable auto summary using no auto-summary.
4. Wait for about 30 seconds (RIP update timer). Check the learnt routes with show ip route and show ip protocols.
   Ping PC0 to PC1 and trace the path.

Device	Interface	IP address	Subnet mask
R1	Gig0/0	192.168.1.1	255.255.255.0
R1	Se0/0/0 (DCE)	10.0.0.1	255.255.255.252
R2	Se0/0/0	10.0.0.2	255.255.255.252
R2	Se0/0/1 (DCE)	10.0.0.5	255.255.255.252
R3	Se0/0/1	10.0.0.6	255.255.255.252
R3	Gig0/0	192.168.2.1	255.255.255.0
PC0	NIC	192.168.1.2	255.255.255.0
PC1	NIC	192.168.2.2	255.255.255.0

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Commands used

Static routing	*RIP*
`R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2`	`R1(config)#router rip`
`R1(config)#ip route 10.0.0.4 255.255.255.252 10.0.0.2`	`R1(config-router)#version 2`
`R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1`	`R1(config-router)#network 192.168.1.0`
`R2(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.6`	`R1(config-router)#network 10.0.0.0`
`R3(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.5`	`R1(config-router)#no auto-summary`
`R3(config)#ip route 10.0.0.0 255.255.255.252 10.0.0.5`	`(similar for R2: 10.0.0.0 and R3: 192.168.2.0)`

DIAGRAM:

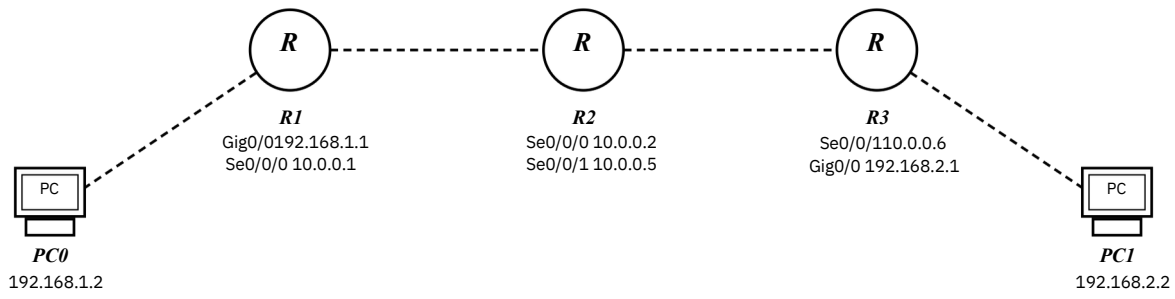
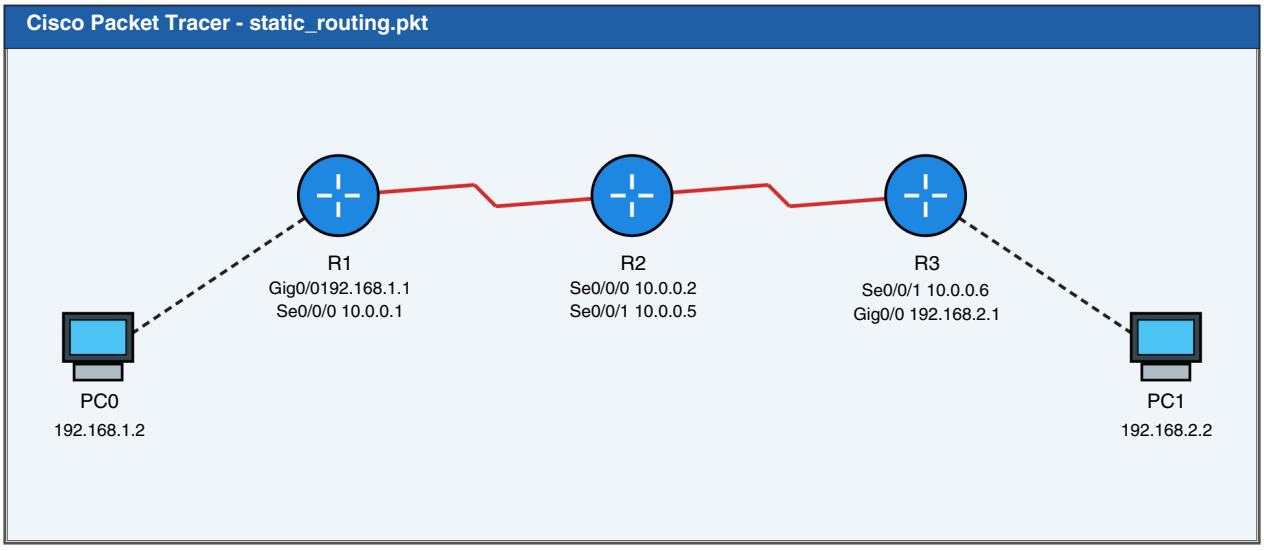


Fig 1: Three routers connected by serial links (dotted lines are serial links)

OUTPUT:

Part (a) - Static routing



Packet Tracer view (red lightning lines are serial links)

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PC0 - Command Prompt (before static routes)

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

R1 - CLI

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
S       10.0.0.4 [1/0] via 10.0.0.2
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.0.2
```

PC0 - Command Prompt (after static routes)

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

C:\>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a maximum of 30 hops:

  1    0ms       0ms       0ms       192.168.1.1
  2    1ms       1ms       0ms       10.0.0.2
  3    2ms       1ms       2ms       10.0.0.6
  4    2ms       2ms       1ms       192.168.2.2

Trace complete.
```

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Part (b) - RIP

R1 - CLI

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#show ip route rip
     10.0.0.0/30 is subnetted, 2 subnets
R       10.0.0.4 [120/1] via 10.0.0.2, 00:00:12, Serial0/0/0
R    192.168.2.0/24 [120/2] via 10.0.0.2, 00:00:12, Serial0/0/0
R1#show ip protocols
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 17 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
  Default version control: send version 2, receive 2
Automatic network summarization is not in effect
 Maximum path: 4
 Routing for Networks:
   10.0.0.0
   192.168.1.0
 Routing Information Sources:
   Gateway         Distance      Last Update
   10.0.0.2              120      00:00:12
 Distance: (default is 120)
```

PC0 - Command Prompt

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=2ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125
Reply from 192.168.2.2: bytes=32 time=1ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	R1	ICMP		0.000	N	0
	Successful	R1	R2	ICMP		0.001	N	1
	Successful	R2	R3	ICMP		0.002	N	2
	Successful	R3	PC1	ICMP		0.003	N	3
	Successful	R1	R2	RIP		30.000	N	4
	Successful	R2	R3	RIP		30.001	N	5

Point	*Static routing*	*RIP*
Route entry	Typed manually by the admin (S)	Learnt automatically from neighbours (R)
Administrative distance	1	120
Updates	None	Every 30 seconds
Link failure	Admin has to change the route	Finds another path automatically
Metric	Not used	Hop count (max 15)
Best for	Small and stable networks	Small to medium networks

RESULT:

Thus static routing and RIP were configured and simulated in Cisco Packet Tracer. Ping between the two remote networks failed without routes and became successful after configuring the static routes and also after configuring RIP.